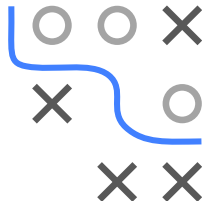
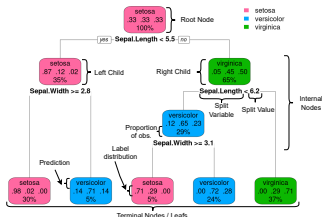


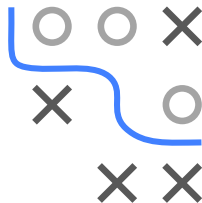
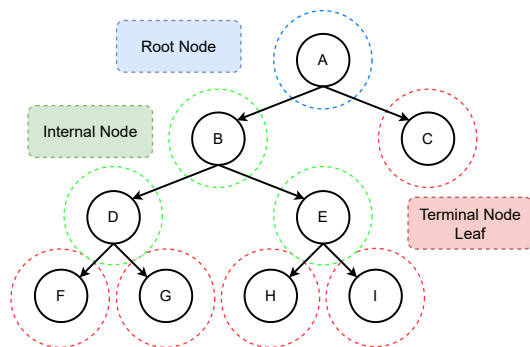
Predictions with CART



- Understand the basic structure of a tree model
- Understand that the basic idea of a tree model is the same for classification and regression
- Understand how the label of a new observation is predicted via CART
- Know hypothesis space of CART

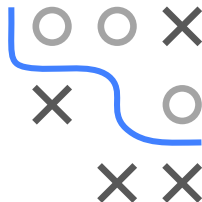
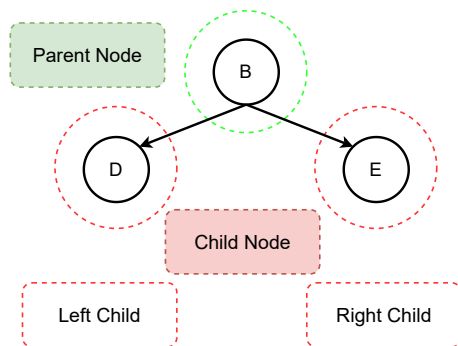


BINARY TREES I



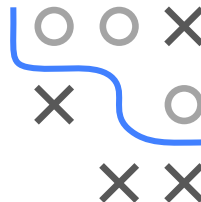
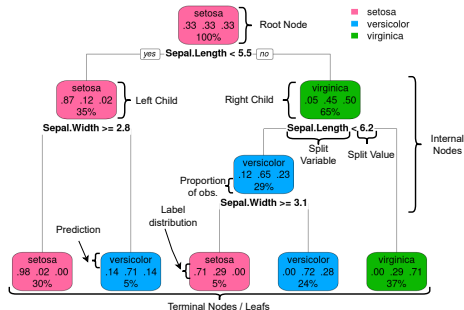
- Binary trees represent a top-down hierarchy with binary splits
- A tree is composed of different node types: a) the root node, b) internal nodes, and c) terminal nodes (also leaves).

BINARY TREES I



- Nodes have relative relationships, they can be:
 - Parent nodes
 - Child nodes
- Root nodes don't have parents – leaves don't have children

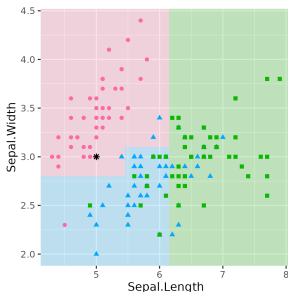
CLASSIFICATION TREES I



- Classification trees use the structure of a binary tree
- Binary splits are constructed top-down in a *data optimal* way
- Each split is a threshold decision for a single feature
- Each node contains the training points which follow its path
- Each leaf contains a constant prediction

1

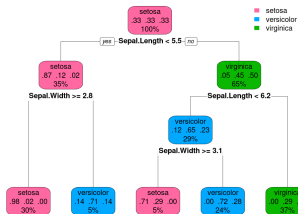
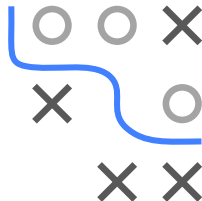
-
- A 3x3 grid with a blue path starting at the top-left corner and ending at the bottom-right corner. The path passes through the middle-right cell. The grid contains several 'X' marks and one 'O' mark.



CART AS A RULE BASED MODEL I

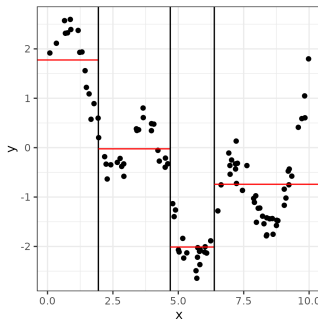
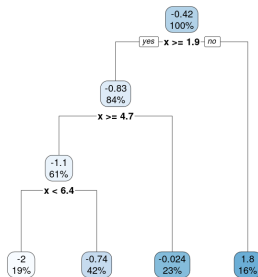
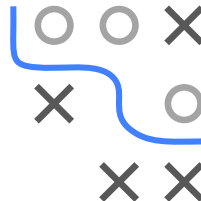
Leaf nodes can be expressed by a set of rules (left to right):

Hard label prediction	Label distribution	Sepal.Width	Sepal.Length
setosa	0.98, 0.02, 0.00	≥ 2.8	< 5.5
versicolor	0.14, 0.71, 0.14	< 2.8	< 5.5
setosa	0.71, 0.29, 0.00	≥ 3.1	$\geq 5.5 \text{ \& } < 6.2$
versicolor	0.00, 0.72, 0.28	< 3.1	$\geq 5.5 \text{ \& } < 6.2$
virginica	0.00, 0.29, 0.71	—	≥ 6.2



REGRESSION TREE MODEL AND PREDICTION I

- Works the same way as for classification
- But predictions in leaf nodes are a numerical scalar



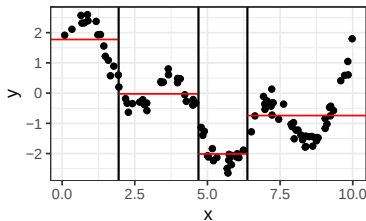
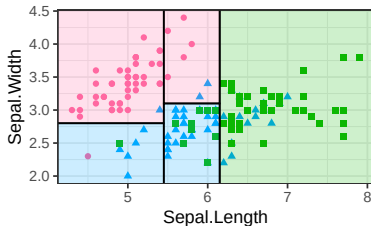
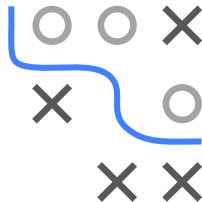
TREE AS AN ADDITIVE MODEL I

Trees divide the feature space $\mathcal{X}$ into **rectangular regions**:

$$f(\mathbf{x}) = \sum_{m=1}^M c_m \mathbb{I}(\mathbf{x} \in Q_m),$$

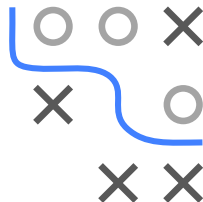
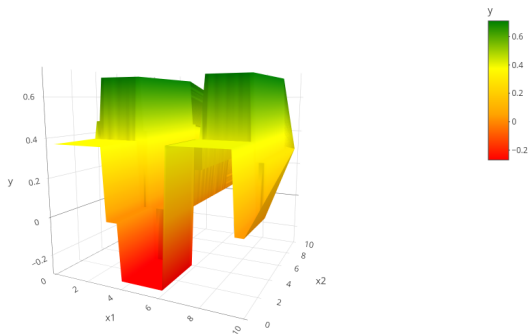
where a tree with M leaf nodes defines M “rectangles” Q_m .

c_m is the predicted numerical response, class label or class distribution in the respective leaf node.



TREE AS AN ADDITIVE MODEL I

A 2D regression example:



(For binary classification with probabilities, 2D surface looks similar.)